Password Door Lock System

In this project, we will create a **Password Door Lock System** using Arduino UNO, a 4x4 Keypad, LCD Display with I2C module, Servo Motor, Buzzer, and Jumper Wires. The system allows door access by entering a correct password via the keypad. If the correct password is entered, the servo motor will rotate and open the door by moving the latch, and a buzzer will beep to confirm access. If the entered password is incorrect, the servo remains locked and a different buzzer sound will indicate the wrong password. The LCD will display important system messages like welcome greetings, password entry status, access granted or denied.

Working Principle

The system operates using a **4-digit password-based security mechanism**. The user enters the password through a keypad, and the system verifies it against a password stored in **EEPROM memory**, allowing the password to remain saved even after power loss.

Correct Password Scenario:

- If the entered password matches the stored password:
    - The buzzer produces a short confirmation beep
    - The system grants access and displays **"ACCESS GRANTED"** and **"WELCOME"** on the LCD
    - The wrong attempt counter is reset to zero
    - The system remains in normal operational mode

Incorrect Password Scenario:

- If the entered password is incorrect:
    - The system notifies the user with a buzzer beep pattern
    - The LCD displays **"WRONG PASSWORD"** and prompts the user to try again
    - The user is allowed up to **three attempts**

Security Lock Mechanism:

- If the password is entered incorrectly **three consecutive times**:
    - The system enters a **security lock state**
    - The buzzer is activated as an alarm and continuously sounds in a timed cycle for **30 seconds**
    - The keypad input is disabled during this period
    - The LCD displays **"SYSTEM LOCKED"** along with a countdown timer showing the remaining lock duration

Recovery Mode:

- After the 30-second lock period:
    - The system automatically resets
    - The wrong attempt counter is cleared
    - The system returns to normal password entry mode

System Interface:
All user interactions, authentication results, and system states are displayed on a **16x2 LCD screen**, ensuring clear real-time feedback to the user. The system provides a reliable and secure embedded authentication mechanism suitable for access control applications.

Materials Required

- Arduino UNO
- Jumper Wires **-**
- LCD Display With I2C Module
- Servo Motor
- 5V Buzzer
- 4x4 Keypad
